

# DAY 13 INTERNSHIP REPORT — DUPLICATE RECORD CHECKS & DATA CLEANING

**Date:** 20-09-2026 | **Tools:** Python (Pandas) & Excel

## 1. Executive Overview

For the Day 13 task, advanced data cleaning techniques were applied using Python's Pandas library to audit a messy customer orders dataset. The primary objectives included identifying and removing duplicate records, standardizing messy string formats into proper numeric types, and logically imputing missing financial values based on existing row data.

## 2. Duplicate Detection & Handling Methodology

The initial dataset was loaded with rows skipped to account for header formatting. An initial pass was made to strip whitespace from column headers and all string-based columns. Explicit string values of 'nan' were standardized to proper null values.

Duplicate checking was split into two categories: **Full-Row Duplicates** and **Order\_ID Duplicates (Exact + Conflicts)**. These findings were cataloged before utilizing the `drop_duplicates()` method to retain only the first instance of each valid order and establish a clean base.

## 3. Data Standardization & Formulaic Imputation

Significant type casting was required for numerical analysis:

| Transformation Step     | Implementation Details                                                                                                                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Financial Data Coercion | Columns like <code>Unit_Price</code> , <code>Total_Amount</code> , and <code>Discount_Pct</code> had string-based commas removed before being converted to numeric floats.                                    |
| Standard Numeric Types  | <code>Customer_Age</code> , <code>Quantity</code> , and <code>Delivery_Days</code> were safely coerced to numeric types.                                                                                      |
| Datetime Parsing        | The <code>Order_Date</code> column was converted using <code>pd.to_datetime</code> .                                                                                                                          |
| Calculated Imputation   | A boolean mask was used to identify rows missing a <code>Total_Amount</code> . The missing values were formulaically imputed using the logic: <code>Quantity * Unit_Price * (1 - Discount_Pct / 100)</code> . |

## 4. Audit Note & Exported Deliverables

To maintain a transparent data lineage, the audit results and the final cleaned data were exported into separate deliverables:

- **Day-13\_Duplicate\_and\_Cleaning\_Report.xlsx:** Contains an audit summary sheet (`Duplicate_Summary`) noting the counts of identified duplicates, and a detailed sheet (`Flagged_Duplicates`) storing the actual duplicate rows for business review.
- **Customer\_Orders\_Cleaned.csv:** The final, fully standardized, and deduplicated dataset ready for downstream analytics.